

Electronic Workshop-2 Project-2 Presentation

*Piezoelectric Vibration Energy Harvesting with AC–DC Rectification
and Voltage Regulation to power Low Power Source nodes*

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Background and Motivation

- ▶ Conventional battery-powered sensor nodes require periodic maintenance and replacement
- ▶ Ambient mechanical energy — footsteps, vibrations — is abundant yet largely unutilised
- ▶ Piezoelectric transducers offer a viable means of harvesting this otherwise wasted energy

Problem Statement

- ▶ Raw piezoelectric output is AC, variable in amplitude, and unsuitable for direct use
- ▶ A complete signal conditioning chain is required to deliver stable, regulated DC power

Objectives

- ▶ Design and implement a full energy harvesting pipeline: AC-DC rectification → impedance buffering → buck conversion
- ▶ Regulate harvested energy to a stable **3.3 V** output
- ▶ Validate the system's ability to power a low-power sensor node from footstep vibrations

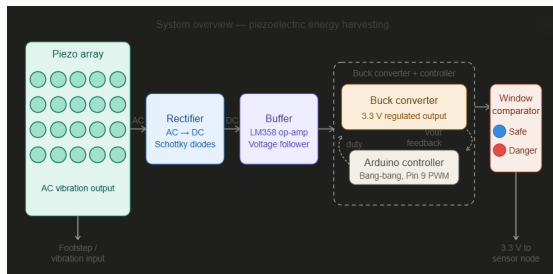


Figure: Block Diagram

Components

- ▶ 5×4 piezoelectric sensor array
- ▶ AC-DC rectifier + LM358 buffer
- ▶ Buck converter + Arduino controller
- ▶ Window comparator with LED indicators

Hardware

- ▶ 5×4 piezo array wired in parallel on a cardboard base
- ▶ Full-wave rectifier using 1N5819 Schottky diodes
- ▶ LM358 op-amp voltage follower as impedance buffer
- ▶ Arduino Uno driving a PWM buck converter at 50 kHz
- ▶ Op-amp window comparator with Blue/Red LED indicators

Software

- ▶ Timer1 configured for fast PWM at 50 kHz ($ICR1 = 319$)
- ▶ Bang-bang controller adjusts duty cycle via OCR1A
- ▶ 16-sample ADC averaging for stable voltage feedback
- ▶ Duty cycle clamped: $DUTY_MIN = 1$, $DUTY_MAX = 150$

- ▶ **Buck converter complexity** — understanding PWM control and relating theoretical principles to practical implementation required significant effort
- ▶ **Soldering sensitivity** — piezo sensors are sensitive to high temperatures; soldering a 4×5 array demanded precision and was time-consuming
- ▶ **Low and intermittent power generation** — inherently weak and mechanically dependent piezo output made obtaining a stable input to downstream stages difficult
- ▶ **Voltage loss in rectification** — diode forward voltage drops in the AC-DC stage reduced the usable output, especially at low input levels
- ▶ **Time and component constraints** — procurement delays limited scope for iterative testing and optimisation
- ▶ **Limited reference material** — few directly applicable references for piezo harvesting with integrated regulation and control

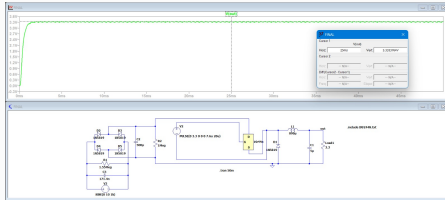
Efficient Voltage Regulation via Buck Converter

- ▶ Voltage regulation is a critical requirement in the energy harvesting pipeline
- ▶ Conventional approaches such as linear voltage regulator ICs or op-amp based regulators dissipate excess voltage as heat, leading to significant power loss
- ▶ A **PWM-controlled buck converter** is employed instead, offering:
 - ▶ Switching efficiency of up to ~85–90% compared to ~40–60% for linear regulators
 - ▶ Minimal heat dissipation — critical for a self-powered, low-energy system

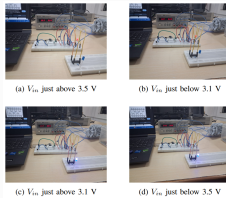
Op-Amp Based Window Comparator for Safety Indication

- ▶ A window comparator circuit is implemented at the output stage
- ▶ Continuously monitors the regulated output and indicates proper or faulty operation.
- ▶ Provides real-time visual indication:
 - ▶ **Blue LED** — output within safe range
 - ▶ **Red LED** — output outside safe range (under/over voltage)

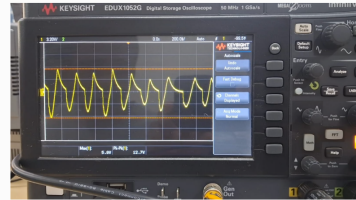
Results & Testing



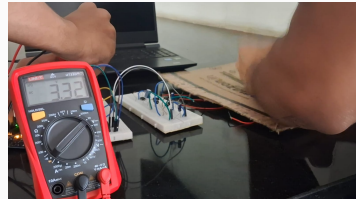
LTspice simulation results



Window Comparator Working



Piezo Array output



Final Circuit Working

Conclusion

- ▶ Successfully designed and implemented a piezoelectric energy harvesting system powered by ambient mechanical vibrations
- ▶ 5×4 sensor array improved voltage and current output; full-wave Schottky rectifier minimized drop to $\approx 0.4 \text{ V}$
- ▶ Buck converter regulated output to a stable **3.3 V** with $\leq 1\%$ deviation using a hysteretic bang-bang Arduino controller
- ▶ Peak output power in the order of hundreds of milliwatts under strong mechanical excitation — sufficient to power low-power IoT sensor nodes
- ▶ Window comparator provides real-time visual safety indication (Blue: safe 3.1–3.5 V, Red: fault)
- ▶ Demonstrates feasibility of piezoelectric harvesting as a battery-free power source for low-power IoT/sensor nodes

References

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